University of Cape Town

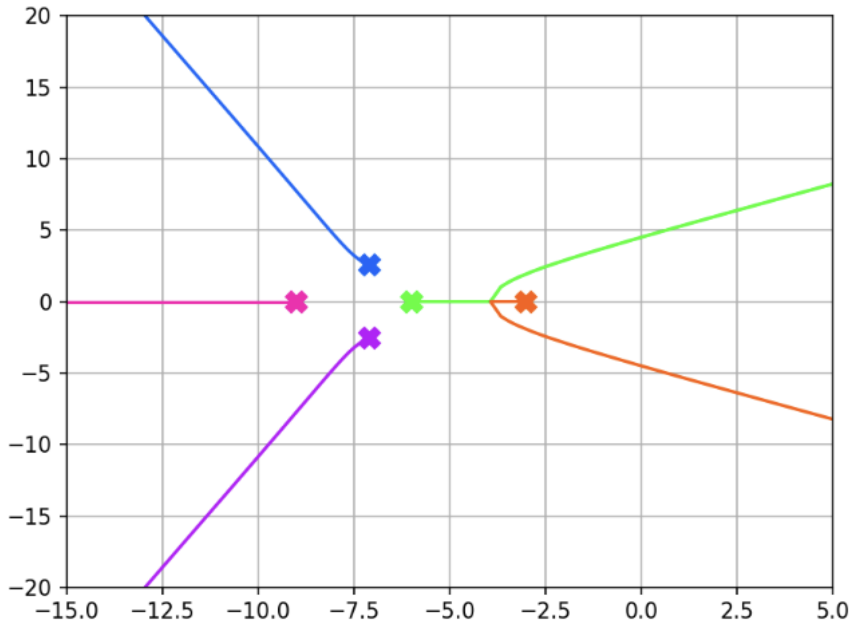
Department of Electrical Engineering

EEE3094S - Control Systems Engineering

Problem Set 4.2: Root Locus Solutions

Question 1

A fifth-order system has the root locus shown below:



True or false: (explain)

a) The system is stable for all gains

b) The system's response will become more oscillatory as the gain increases

c) One of the asymptotes has an angle of 108 degrees

d) Within the range of gains for which the system is stable, the response will settle faster as the gain increases.

Solution:
a) False, the orange and green tracks become unstable as gain increases

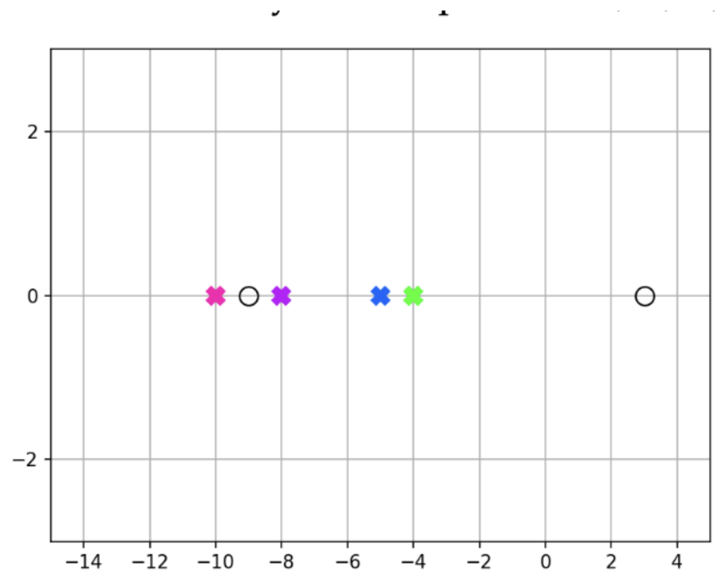
b) True, as gain increases the closed loop poles move further from the real axis resulting in oscillatory behaviour.

c) True, there are 5 poles. $180/5 = 36$. The asymptotes depart on odd multiples: ± 36, ± 108, ± 180

d) False, the poles closer to the imaginary axis will become the dominant poles. The system will initially speed up as the orange pole moves left and slow down as the orange green pair moves right.

Question 2

A fourth-order system has poles at -10, -8, -5, and -4, and zeros at -9 and 3:



a) Will the system be stable in closed-loop for any gain?

b) How many asymptotes will the root locus have? Find their angles and point of intersection on the real axis.

c) Sketch a rough root locus plot for the system (you only need to show which sections of the real axis are included and the trajectories of the poles along the asymptotes. Specific values of breakaway points, angles of departure, etc. are not required).

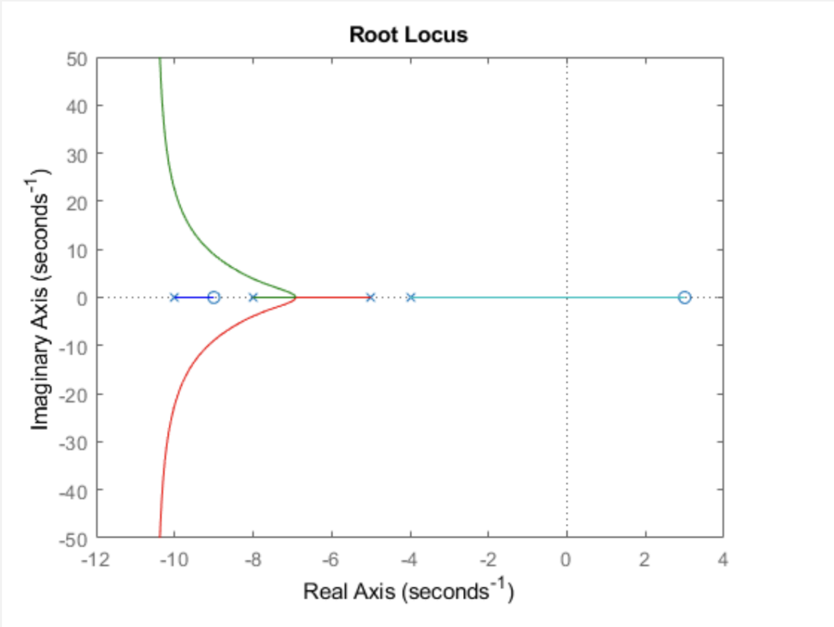
d) Based on your plot, answer the following true or false questions about the system (with reasons):

   i) A first-order model is an acceptable representation of the system's behaviour at some gains.
   ii) All the system's modes have a finite maximum frequency.
   iii) All the system's modes have a finite minimum settling time.
   iv) Adding a zero at -1 would ensure that the system will be stable for all gains.

Solution:
a) No, there is a zero in the right half plane

b) The asymptote angles are given by $\frac{\pi}{n-m} = \frac{\pi}{2} = \pm 90$

c)



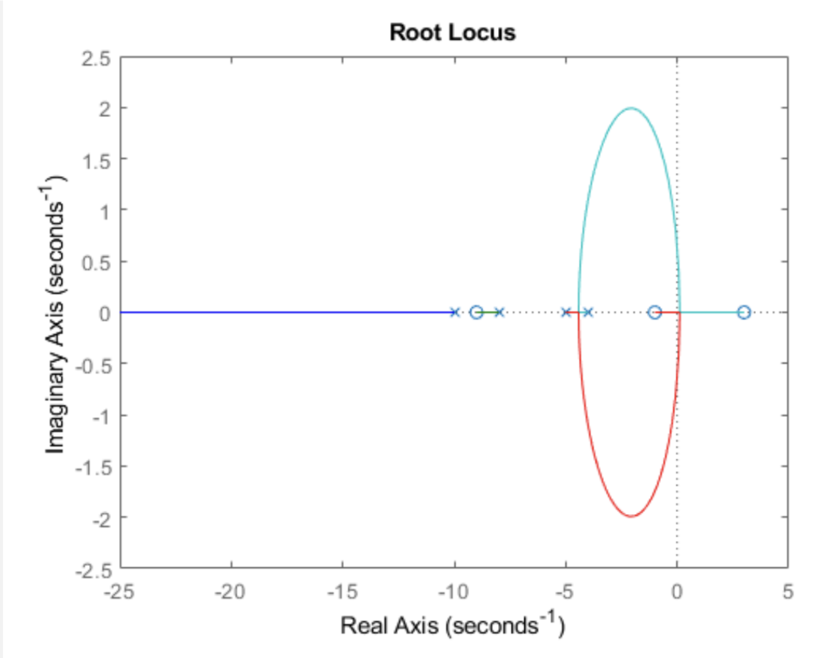
d)

i) True, the pole at -4 will be the dominant response as gain increases

ii) False, two poles go to infinity on the complex axis. Therefore, there is no maximum frequency.

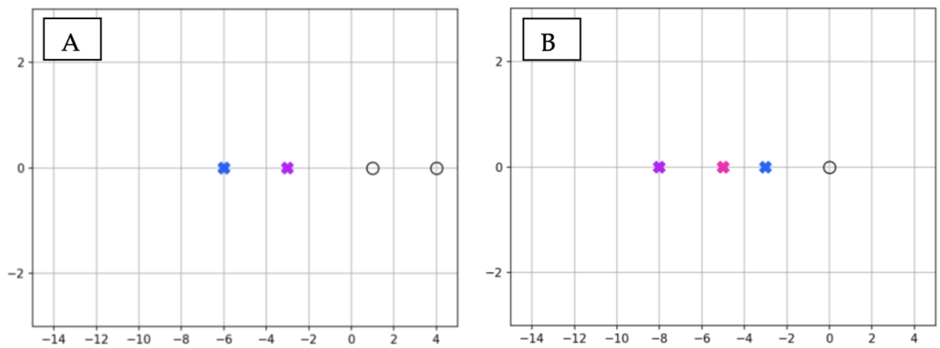
iii) True, the minimum settling time is given by the slowest pole and none go to infinity on the real axis.

iv) False, the zero at 3 would still result in an unstable system


Root Locus
Imaginary Axis (seconds^{-1})
Real Axis (seconds^{-1})
2.5
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
-2.5
-25
-20
-15
-10
-5
0
5

Question 3

Consider the pole-zero plots of the following third-order systems (A has a repeating pole at -6):



By sketching rough root locus plots of these systems, determine:

a) Which root locus exists on the real axis at -4?

b) Which system has a maximum gain?

c) Which system's response has a finite maximum frequency?

d) Which system can be approximated by a first-order model at some gains?

e) Which system can be approximated by a second-order model at some gains?

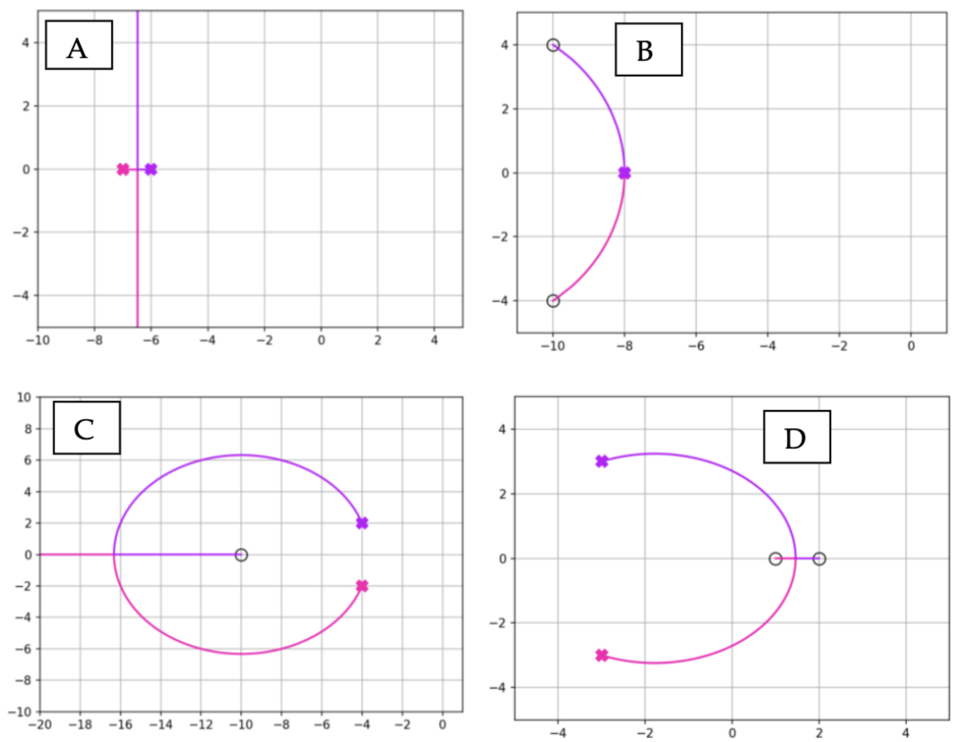
f) Which system's response will settle more quickly as the gain increases?

(answers can be one, or both, or neither)
Bonus: sketch the complete root locus for these systems, including all values of interest.

Solution:
a) A
b) A, maximum for stability
c) B, the poles will breakaway and diverge
d) B
e) A
f) Neither

Question 4

Complete the following root locus sketches by filling in all values of interest (asymptotes, breakaway/-in points, imaginary axis crossings, angles of departure/arrival). The real and imaginary parts of all poles and zeros are integers.



Solution:
System A:

- Asymptotes: ± 90
- Breakaway: -6.5
- Imaginary axis crossings: None
- Departure and arrival angles: 90 departure, No arrival angles

System B:

- Asymptotes: No asymptotes, equal number of poles and zeros
- Breakaway/in: -8
- Axis crossings: None
- Departure angle: $\frac{\pi}{2}$ rad, 90
- Arrival angle: 0.643 rad, 36.8

System C:

- Asymptotes: 180
- Breakin: $\lambda^2 + 20\lambda + 60$, Break in at -16.3
- Axis crossings: None
- Departure angle: 1.89 rad, 108.3
- Arrival angle: 0

System D:

- Asymptotes: No asymptotes, equal number of poles and zeros
- Breakin: 1.46
- Axis crossings: The transfer function is $\frac{s^2-3s+2}{s^2+6s+18}$

Which results in a closed loop characteristic equation of: $s^2 + 6s + 18 + k(s^2 - 3s + 2)$

$$\begin{array}{c|cc} s^2 & k+1 & 2k+18 \\ s^1 & -3k+6 & 0 \\ s^0 & 2k+18 & \end{array}$$

Therefore $k \leq 2$ for the system to be stable
Substituting $k = 2$ into the characteristic equation produces $3s^2 + 22$
Therefore the axis crossing is at $\pm 2.7j$

- Departure angles: 0.39 rad, 22.3
- Arrival angles: 0 and 180

Question 5

A spring-mass-damper system has a mass $m = 1$, damping coefficient $b = 3$ and spring constant $k = 6$. The transfer function of the system is

$$\frac{1}{ms^2 + bs + k}$$

a) Sketch a root locus showing how variation in the i) mass ii) damping coefficient and iii) spring constant affects the pole positions. (Keep the other parameters constant at their nominal values in each case).

b) If each of these parameters is subject to a 5% tolerance, calculate the (maximum) root sensitivity within this range of the nominal value. Which parameter's fluctuations will affect the system's response the most?

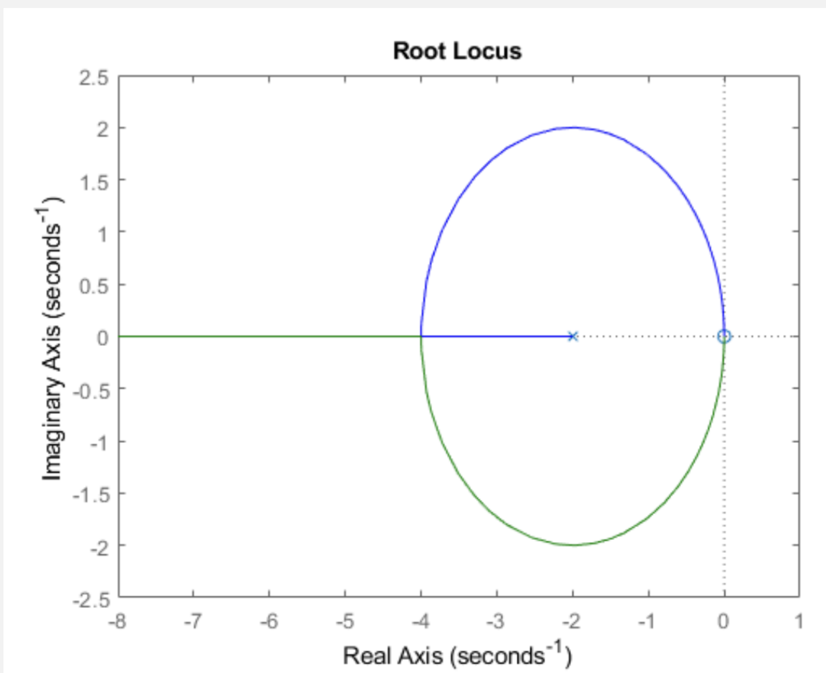
Solution:

a)

i. Mass variation: Characteristic equation: $ms^2 + 3s + 6 = 0$

Changing the equation to make m the gain: $1 + \frac{ms^2}{3s+6} = 0$

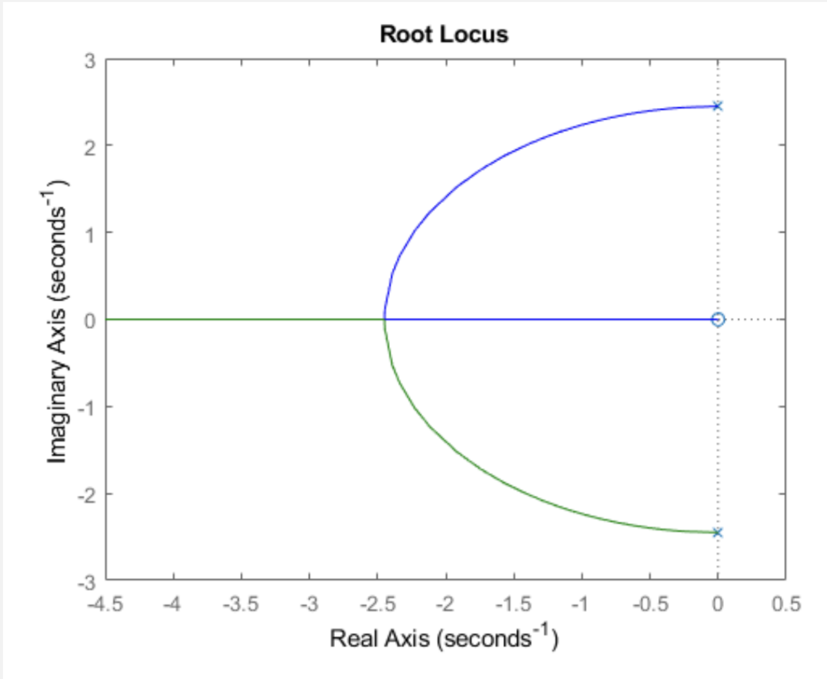
The transfer function has a pole at -2 and two zeros at the origin. As mass increases a pole will come from infinity to form a complex pair. As mass is increased the system will become oscillatory and the response will slow down. Increasing mass will not result in instability.



ii. Damping coefficient variation: Characteristic equation: $s^2 + bs + 6 = 0$

Changing the equation to make b the gain: $1 + \frac{bs}{s^2+6} = 0$

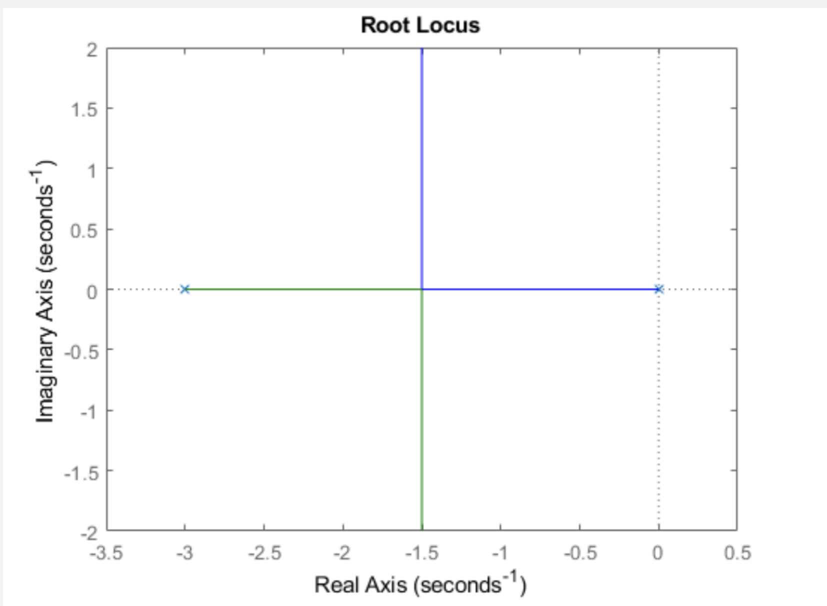
There are two poles at $\pm\sqrt{6}j$ and a zero at the origin. At low damping factors the system is marginally stable. Increasing the damping factor reduces oscillation but slows down the system.



iii. Spring constant variation: Characteristic equation: $s^2 + 3s + k = 0$

Changing the equation to make k the gain: $1 + \frac{k}{s^2+3s} = 0$

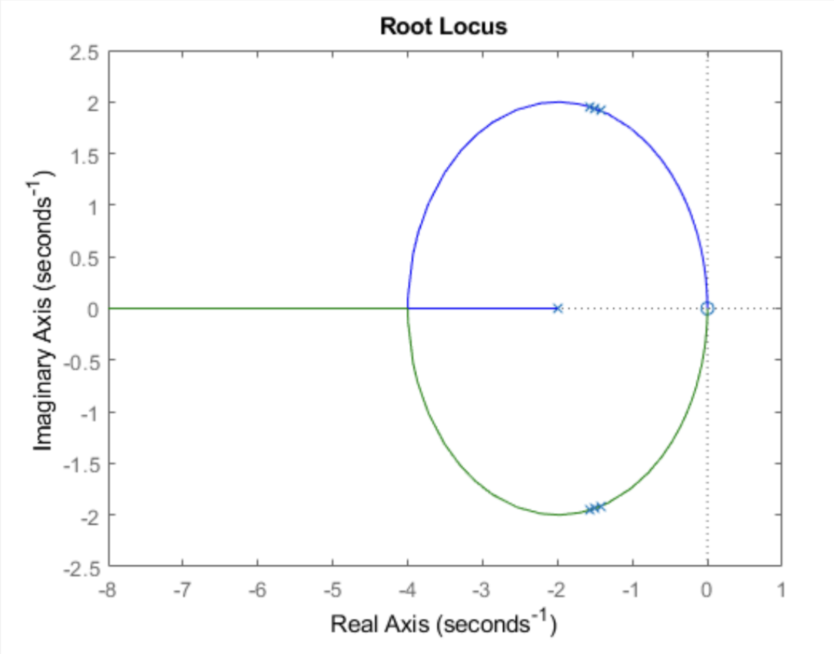
There are poles at the origin and -3. The poles will move towards each other before breaking away at 90. The system will become more oscillatory with increasing spring constant but will not become unstable.



b) For $0.95 \leq m \leq 1.05$ the poles are at:

$$-1.58 \pm 1.96j \quad (1)$$
$$-1.5 \pm 1.94j \quad (2)$$
$$-1.43 \pm 1.92j \quad (3)$$

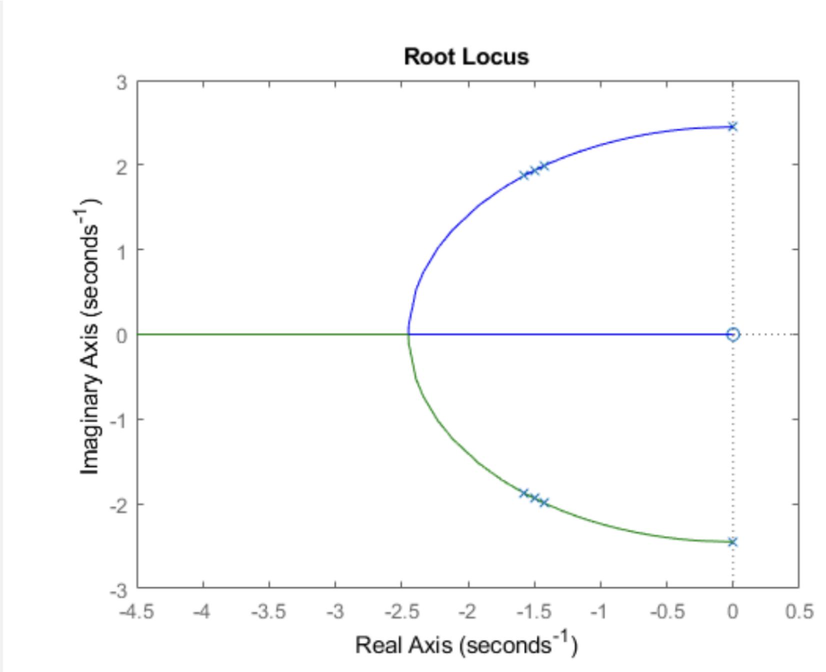
Sensitivity: $\frac{\delta p/p}{\delta m/m} = \frac{-0.08-0.02j}{-1.5+1.94j} \cdot \frac{0.01}{1} = 0.67\angle -2.47$



For $2.85 \leq b \leq 3.15$ the poles are at:

$$-1.43 \pm 1.99j \quad (4)$$
$$-1.5 \pm 1.94j \quad (5)$$
$$-1.58 \pm 1.88j \quad (6)$$

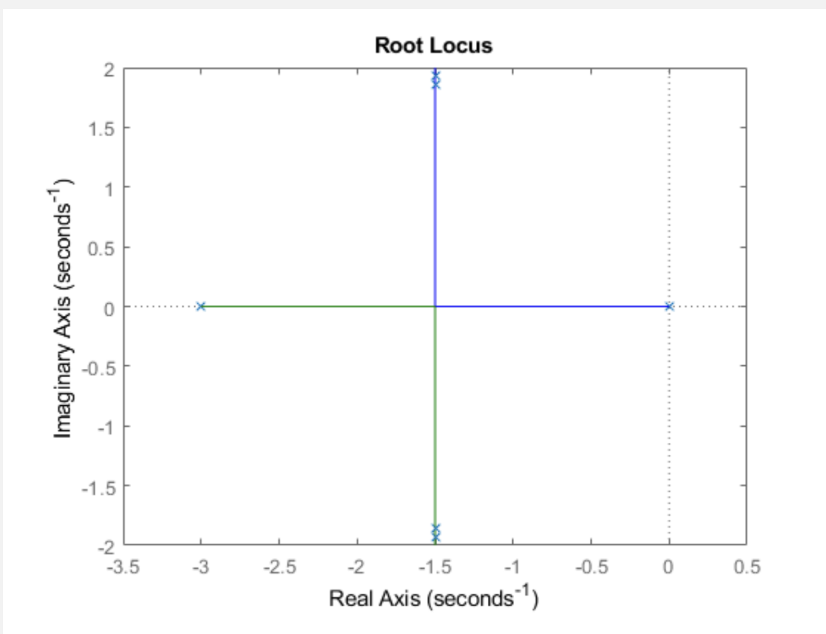
Sensitivity: $\frac{\delta p/p}{\delta b/b} = \frac{-0.08-0.06j}{-1.5+1.94j} \cdot \frac{0.15}{3} = 0.81\angle 1.56$



For $5.7 \leq k \leq 6.3$ the poles are at:

$$-1.5 \pm 1.85j \tag{7}$$
$$-1.5 \pm 1.94j \tag{8}$$
$$-1.5 \pm 2.01j \tag{9}$$

Sensitivity: $\frac{\delta p/p}{\delta k/k} = \frac{0.09j}{1.94j} \cdot \frac{0.3}{6} = 0.92$



Conclusion: The spring constant k has the highest sensitivity (0.92), so its fluctuations will affect the system's response the most.